

PYTHON FUNDAMENTALS

Python:

Python is an interpreted, high level & general purpose programming language.

It was created by Guido Van Rossum in 1989 and first released in 1991.

Whenever we run python program, python compiler implicitly converts source code into byte code, python virtual machine converts byte code into machine code and machine code executed to get the output.

Both compiler & interpreter are called translation softwares.

Differences between compiler & interpreter:

Compiler converts the whole program at a time whereas interpreter converts line by line.

Compiler will produce the file whereas interpreter does not produce the file.

Python is a platform independent.

Programs written in python programming language can be executed on any platform.

Platform:

It can be a software or hardware environment in which program runs.

Applications of Python:

- 1) Artificial Intelligence Applications
- 2) Machine Learning Applications
- 3) Data Science Applications
- 4) Data Visualization Applications
- 5) Web Scrapping Applications
- 6) Scientific & Numeric Applications
- 7) Internet Of Things(IOT) Applications
- 8) Image Processing Applications

9) Web Applications

10) Enterprise Applications, .. etc.,

Identifiers:

Identifier is a word and it is used to identify variables, functions, methods, classes, packages, .. etc.,

Identifier can be a variable name, function name, method name, class name, package name, .. etc.,

Rules to declare an identifier:

- 1) It can be formed by using alphabets(A to Z & a to z), digits(0 to 9), underscore symbol(_).
- 2) It must begins with alphabet, underscore symbol(_).
- 3) The length of the identifier is not limited.
- 4) It should not contain special symbols other than underscore symbol.
- 5) It should not contain white space characters(Space bar, tab & enter keys).

Keywords:

A set of words reserved by language itself and those words are called keywords.

Examples:

and, or, not, if elif, else, while, for, break, continue, def, del, in, is, try, with, except, finally, ..etc.,

All keywords must be written in lowercase letters only except True, False & None.

There are at present total 36 keywords in Python.

Note: Keyword cannot be used as an identifier.

Literals:

A literal is a source code representation of a fixed value.

In python, literals are divided into 4 categories:

1) Numeric Literals:

2) String Literals:**3) Boolean Literals:****4) Special Literals:****1) Numeric Literals:** Numeric literals are divided into 3 categories.i) Integer Literals:

5, 88, 234, 0, -2, -13, -483, .. etc.,

ii) Floating Point Literals:

5.2, 2.832, 0.088, -2.456, .. etc.,

iii) Imaginary Literals:

3+2j, 2+4j, 9+4j, -2+4j, -9-2j, .. etc.,

2) String Literals: String literals are divided into 2 categories.

i) Single Line String Literals: Single line string literals can be in single quotes, double quotes, triple single quotes & triple double quotes.

Examples:

'hello', "hello", '''hello''', """hello""", .. etc.,

ii) Multiline String Literals: Multiline string literals can be in triple single quotes & triple double quotes.

Examples:

'''Welcome

To Python''',

"""Welcome To

Python""", .. etc.,

3) Boolean Literals:

True & False

4) Special Literals:

None

Data Types:

A data type that determines what value variable can hold and what are the operations can performed on variables.

In python, data types are divided into 3 categories:

- 1) Fundamental Types
- 2) Collection Types
- 3) Special Types

1) Fundamental Data Types:

There are five fundamental data types.

- 1) int
- 2) float
- 3) complex
- 4) str
- 5) bool

2) Collection Data Types:

There are five collection data types.

- 1) list
- 2) tuple
- 3) set
- 4) frozenset
- 5) dict

3) Special Data Types:

There are four special data types.

- 1) bytes
- 2) bytearray
- 3) range
- 4) NoneType

Variables:

A variable is a container that contains data.

Examples:

a=10

b=5.2

c=3+4j

```
d="Hello"
```

```
e=True
```

print() function:

It is used to display data on monitor.

Examples:

```
a=10
```

```
print(a)
```

```
b=20.5
```

```
print(b)
```

type() function:

It returns type of an object.

Examples:

```
a=10
```

```
type(a)
```

```
b=10.25
```

```
type(b)
```

By

Mr. Venatesh Mansani

Naresh i Technologies